

Notes: Maksimovic and Phillips (JF, 2001)

The Market for Corporate Assets: Who Engages in Mergers and Asset Sales and Are There Efficiency Gains?

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1 One-sentence summary

Maksimovic and Phillips use plant-level Census data to show that the U.S. market for corporate assets reallocates plants in ways largely consistent with efficient neoclassical reallocation: assets are more likely to be sold when sellers are relatively less productive and buyers are relatively more productive, especially for partial-firm asset sales.

2 Big picture

2.1 Research question

The paper asks whether mergers, acquisitions, and partial asset sales improve the allocation of corporate assets, or whether they mainly reflect agency problems such as empire building and inefficient diversification.

2.2 Why the question matters

Most merger studies use firm-level stock returns or accounting data. This paper instead studies the actual operating assets being traded: manufacturing plants. The plant-level view matters because many transactions are partial-firm sales of plants or divisions rather than whole-firm mergers.

2.3 Core finding

There is an active market for corporate assets. Around 3.9 percent of large manufacturing plants change ownership annually from 1974 to 1992, with rates much higher in expansions. Reallocations are related to productivity, firm organization, and industry demand.

2.4 Economic mechanism

Firms tend to sell assets when their relative productivity in those assets is low and demand conditions make reallocation attractive. Buyers tend to acquire assets when they are productive and growing in the relevant industries. Plants sold by less efficient owners to more efficient owners often exhibit productivity improvements.

2.5 Big takeaway

The market for corporate assets is not simply a symptom of managerial agency problems. It also functions as a mechanism for reallocating real assets toward higher-valued uses.

3 Incremental contribution of the paper

3.1 A plant-level view of corporate asset markets

Earlier merger studies often focus on firm-level stock returns. This paper uses the Longitudinal Research Database to track individual plants, allowing the authors to observe which physical assets move and how their productivity changes after transactions.

3.2 Mergers and asset sales in one framework

The paper treats mergers, acquisitions, full-segment sales, and partial-segment sales as components of one market for corporate assets. This is important because partial asset sales make up a large share of transactions but are underrepresented in traditional merger data.

3.3 Distinguishing efficiency from agency explanations

The paper tests whether transactions follow patterns predicted by efficient reallocation: low-productivity sellers, high-productivity buyers, and post-transaction productivity gains. This goes beyond event-study evidence that transactions create or destroy shareholder value.

3.4 Firm organization as a determinant of transactions

The paper shows that conglomerates and single-segment firms behave differently. Conglomerates are more likely to sell peripheral divisions and less likely to sell main divisions, consistent with an internal comparative advantage over assets.

3.5 Contribution to theories of diversification

The results provide evidence that diversified firms do not uniformly destroy value. They may own some assets efficiently while disposing of assets where they lack comparative advantage. This complements the literature on diversification discounts and internal capital markets.

4 Data and measurement

4.1 Plant-level transaction data

The analysis uses plant-level data from the Census Bureau’s Longitudinal Research Database and Annual Survey of Manufactures. The data track manufacturing plants over time and identify changes in ownership through mergers, acquisitions, and asset sales.

4.2 Transaction categories

The paper classifies reallocations into:

- mergers and acquisitions, where all assets of a selling firm are acquired;
- full-segment sales, where a firm sells an entire business segment;
- partial-segment sales, where a firm sells some plants inside a segment while remaining active in that segment.

Interpretation: This classification lets the paper separate whole-firm transactions from the partial asset market, where agency and efficiency explanations can look quite different.

4.3 Productivity measurement

Plant productivity is measured using total factor productivity estimated at the plant level. The paper also uses robustness measures such as value added per worker and cash flow.

Interpretation: Plant-level productivity allows the authors to ask whether a plant is more productive under one owner than another. This is the core variable that lets the paper test efficient reallocation.

4.4 Firm organization and segment rank

Firms can operate in one or many industry segments. For multi-segment firms, segments are ranked by real value of shipments. The largest segment is the main division; smaller segments are peripheral divisions.

Interpretation: Segment rank captures a firm’s likely core competency. If firms sell peripheral segments more often, that is consistent with comparative-advantage logic.

4.5 Industry demand

Industry demand is measured using detrended aggregate industrial production and industry-level demand changes. Expansion and recession periods are defined relative to long-run trends in industry output.

Interpretation: Demand shocks matter because assets are more valuable when downstream demand is high. The paper tests whether reallocation responds to these shocks.

5 Model and empirical specifications

5.1 Neoclassical reallocation hypothesis

The efficient-reallocation hypothesis predicts that assets move when another owner can operate them more productively or when demand makes the asset more valuable under a different firm.

$$Pr(Sale_{pft} = 1) = F(Productivity_{pft-1}, SegmentProductivity_{ft-1}, Demand_{jt}, FirmOrganization_{ft-1}, Controls). \quad (1)$$

Interpretation: A negative coefficient on plant or segment productivity in a sale regression means low-productivity assets are more likely to be sold. A positive coefficient on buyer productivity in acquisition regressions means productive firms are more likely to buy.

5.2 Partial-firm sales regression

The dependent variable equals one if a plant is sold while the seller remains in operation. The key regressors include plant productivity, segment productivity, aggregate industrial production, industry sensitivity to industrial production, firm size, number of plants, and segment rank.

5.3 Merger and whole-firm acquisition regressions

The whole-firm regressions model the probability that a firm is sold or buys another firm. These regressions distinguish single-segment firms from multi-segment conglomerates and classify conglomerate purchases as focus-increasing or focus-decreasing.

5.4 Post-transaction productivity regressions

The productivity regressions compare changes in productivity around the transaction window from year -1 to year +2:

$$\Delta TFP_p = TFP_{p,t+2} - TFP_{p,t-1}. \quad (2)$$

These changes are compared across buyer-seller types and between cases where buyer productivity exceeds or falls short of seller productivity.

6 Identification and empirical design

6.1 What identifies efficient reallocation

The paper identifies efficient reallocation by looking for three patterns:

1. low-productivity assets are more likely to be sold;
2. high-productivity firms are more likely to buy assets;
3. transacted assets improve productivity when they move from less productive to more productive owners.

6.2 Why plant-level data are crucial

Firm-level data cannot distinguish a productivity gain at the acquired asset from portfolio reshuffling or accounting changes. Plant-level data make it possible to measure performance of the same physical asset before and after ownership changes.

6.3 Agency interpretation versus efficiency interpretation

The agency view predicts that acquisitions may reflect empire building and inefficient diversification. The efficiency view predicts asset movements along comparative-advantage lines. The paper's evidence is more consistent with the efficiency view, especially for partial-firm asset sales.

6.4 Main threats

The main concerns are selection into transactions, measurement error in TFP, unobserved local demand shocks, and survivorship. The paper addresses these with alternative efficiency measures, industry and year adjustments, robustness regressions, and productivity changes around the transaction.

7 Tables and figures

7.1 Table I: Asset Reallocations: Summary Statistics

Read: Table I documents the size and composition of the market for corporate assets.

- 35,291 plants are reallocated from 1974 to 1992.
- The average annual reallocation rate is 3.89 percent.
- Mergers and acquisitions represent about half of reallocated plants.
- Partial-segment and full-segment asset sales together are economically large.

Economic message: The market for assets is large enough to be a major channel of resource reallocation, not a rare corporate event.

Table I
Asset Reallocations: Summary Statistics

Sample characteristics for asset reallocations for the years 1974 to 1992. Reallocations include partial firm asset sales and sales through mergers and acquisitions in which the selling firm disposes of all its assets. Plant-level data is obtained from the Annual Survey of Manufactures from the Bureau of the Census, U.S. Department of Commerce. The average annual percentage of plants reallocated is the number of plants reallocated in a given year divided by the total plants, averaged over all years. The percentage sold to buyer inside industry excludes plant sales to buyers with no existing plants in manufacturing prior to the purchase. Average plant sizes are the total value of shipments in thousands of dollars for each plant deflated by three-digit SIC code deflators from the Bureau of Economic Analysis.

	Sample of Firms			
	Total	Mergers and Acquisitions	Asset Sales	
			Full Segment	Partial Segment
Panel A: Reallocation rates across and within industries				
Full period: 1974 to 1992				
Number of plants reallocated	35,291	17,720	8,556	9,015
Average annual % of plants reallocated	3.89%	1.95% ^{a,c}	0.95%	0.99%
% Plants sold to buyer inside industry				
Same three-digit SIC code	56.8%	54.1% ^{b,c}	55.5%	63.1%
Same four-digit SIC code	47.7%	44.9% ^{a,c}	47.9%	53.0%
Average plant size	\$30,332	\$28,435	\$30,916	\$33,906
(Real \$ in thousands, value of shipments)	\$35,790	\$34,569	\$36,440	\$37,574
Average industry plant size				
(Real \$ in thousands, value of shipments)				

(a) Table I, part 1: overall reallocation rates and industry-related summary statistics.

7.2 Table II: Asset reallocations: Summary Statistics; Table III: Buyer and Seller Characteristics

Read: Table II shows that reallocations are procyclical and related to industry capacity utilization and growth. Table III describes buyers and sellers.

- Reallocations are higher in expansion years than recession years.
- Mergers and full-segment sales are especially cyclical.
- Buyers tend to be larger and more diversified than sellers.
- Buyers are often active in the seller’s industry or related industries.

Economic message: The market for assets is responsive to macro and industry conditions, and buyers are not random. Buyers are frequently firms with operational links to the assets they acquire.

Table II Asset reallocations: Summary Statistics				
Sample characteristics for asset reallocations for the years 1974 to 1992. Reallocations include partial firm asset sales and sales through takeovers and mergers in which the selling firm disposes of all its assets. Plant-level data is obtained from the Annual Survey of Manufactures from the Bureau of the Census, U.S. Department of Commerce. Recession (expansion) years are the three years classified as having the largest decline (expansion) in the aggregate real value of industrial production. Industry capacity utilization quartiles are yearly quartiles based on the rates reported by the Department of the Census. Long-run industry growth/decline quartiles are calculated using growth rates for aggregate industry shipments over a 16-year period, with beginning and ending periods representing three-year averages for 1974 to 1976 and 1990 to 1992.				
	Sample of Firms			
	Total	Mergers and Takeovers	Asset Sales	
			Full Division	Partial Division
Transactions by aggregate economy conditions				
Recession years (1981, 1982, 1991)				
Average % reallocated (total number)	3.57% (5,148)	2.16% (3,112)	0.70% (1,003)	0.72% (1,033)
Expansion years (1986, 1987, 1988)				
Average % reallocated (total number)	6.19% (8,989)	2.69% (3,904)	1.73% (2,509)	1.77% (2,576)
Indeterminate Years	3.21%		0.70%	0.78%
Transactions by industry capacity utilization				
Low industry capacity utilization (bottom quartile)				
Average % reallocated (total number)	3.86% (8,618)	1.90% (4,244)	0.99% (2,210)	0.97% (2,164)
High industry capacity utilization (top quartile)				
Average % reallocated (total number)	3.69% (8,413)	1.92% (4,375)	0.87% (1,977)	0.90% (2,061)
Transactions by long-run industry growth/decline				
Quartile 1: Declining industry growth				
Average % reallocated (total number)	4.01% (6,290)	1.95% (3,058)	1.09% (1,707)	0.97% (1,525)
Quartile 2				
Average % reallocated (total number)	3.86% (5,250)	1.96% (2,666)	1.05% (1,425)	0.85% (1,160)
Quartile 3				
Average % reallocated (total number)	3.52% (10,008)	1.80% (5,131)	0.88% (2,505)	0.83% (2,372)
Quartile 4: High industry growth				
Average % reallocated (total number)	4.03% (15,746)	2.01% (7,870)	0.87% (3,405)	1.14% (4,471)

(a) Table II: asset reallocations by aggregate and industry conditions.

	Sample of Firms			
	Total	Mergers and Acquisitions	Asset Sales	
			Full Segment	Partial Segment
			Panel B: Reallocation rates by 2-digit SIC code	
Industry reallocations: Highest quartile				
SIC Code: 30, Rubber and plastic products	4.72% (2167)	2.03% (974)	1.33% (577)	1.36% (616)
Yearly percentage plant sales (number of sales)				
SIC Code: 36, Electronics/communications	4.61% (3037)	2.12% (1397)	1.28% (840)	1.21% (800)
Yearly percentage plant sales (number of sales)				
SIC Code: 33, Primary metal products	4.49% (1663)	1.95% (732)	1.33% (487)	1.20% (444)
Yearly percentage plant sales (number of sales)				
SIC Code: 38, Scientific/optical equipment	4.38% (1113)	2.22% (570)	1.32% (332)	0.83% (211)
Yearly percentage plant sales (number of sales)				
SIC Code: 20, Processed food products	4.24% (2358)	2.23% (2358)	0.58% (603)	1.40% (1491)
Yearly percentage plant sales (number of sales)				
Industry reallocations: Lowest quartile				
SIC Code: 29, Oil refining	3.21% (307)	2.13% (307)	0.51% (71)	0.58% (82)
Yearly percentage plant sales (number of sales)				
SIC Code: 24, Lumber and wood products	3.04% (1436)	1.54% (743)	0.59% (273)	0.90% (420)
Yearly percentage plant sales (number of sales)				
SIC Code: 27, Printing and publishing	2.99% (1104)	1.84% (1104)	0.57% (331)	0.57% (344)
Yearly percentage plant sales (number of sales)				
SIC Code: 23, Clothes and apparel	2.95% (1495)	1.78% (893)	0.55% (275)	0.62% (327)
Yearly percentage plant sales (number of sales)				
SIC Code: 31, Leather and leather products	2.77% (253)	1.63% (143)	0.68% (63)	0.46% (47)
Yearly percentage plant sales (number of sales)				
* ^a Test of whether the proportion of mergers is significantly different from the proportion of full-segment sales at the 1 percent and 5 percent levels, respectively, using a two-tailed test.				
Test of whether the proportion of mergers is significantly different from the proportion of combined full- and partial-segment sales at the percent level using a two-tailed test.				

(b) Table I, part 2: reallocations by high- and low-reallocation industries.

Table III Buyer and Seller Characteristics					
Sample characteristics of purchasing firms prior to asset purchases for the years 1974 to 1992. Data is aggregated to firm-level from individual manufacturing plants. Plant-level data is from the Annual Survey of Manufactures from the Bureau of the Census, U.S. Department of Commerce. Buyers without any prior manufacturing plants (foreign buyers, outside manufacturing buyers) are excluded as prepurchase characteristics can not be calculated. Average buyer and seller size is the average value of total shipments deflated by industry price deflators from the Bureau of Economic Analysis. 1992 was the last year available at the time the study was conducted.					
	Sample of Firms				
	Asset Sales		Mergers and Takeovers		
	Full Division	Partial Division	Firms with Plants = 1	Firms with Plants > 1	Multiple Industry Firm (Three-digit SIC Code)
Seller characteristics prior to sale					
Full period: 1974 to 1992					
Number of selling firms	3,774	4,205	9,480	2,204	1,224
Average number of plants	23.72	31.48	1.00	5.15	7.0
Average seller size (millions \$)	1,928	1,849	22	176	239
Average number of three-digit industries	7.56	7.77	1.00	3.21	3.17
Average number of four-digit industries	9.58	10.56	1.00	2.53	3.61
% of plants sold by seller in its primary line of business					
Seller's primary three-digit line(s) ^a	28.2%	50.8%	100.0%	70.6%	62.2%
Seller's primary four-digit line(s) ^a	25.6%	36.3%	100.0%	61.6%	54.9%
Buyer characteristics prior to purchase					
Full period: 1974 to 1992					
Number of buyers	2,267	2,755	1,648	4,530	3,827
Average number of plants	23.50	23.38	1.00	22.33	36.80
Average buyer size (millions \$)	1,307	1,357	27.9	1,157	1,410
Average number of three-digit industries	6.24	5.75	1.00	6.00	7.24
Average number of four-digit industries	8.21	7.58	1.00	7.87	9.51
% of plants bought that are in buyer's home industry					
At the three-digit SIC code	53.8%	63.1%	59.1%	30.8%	26.3%
At the four-digit SIC code	47.9%	55.2%	49.9%	21.0%	17.4%
If the seller (buyer) produces in multiple industries, the seller's (buyer's) home industries are those industries that have at least 20 percent of the firm's sales.					

(b) Table III: buyer and seller characteristics.

7.3 Figure 1 and Figure 2: Time-series patterns in ownership changes

Read: Figure 1 plots the annual percentage of plants changing ownership and aggregate industrial production. Figure 2 separates ownership changes by expansion and recession industries.

- Reallocation rates rise strongly in expansions.
- The rate peaks during the late 1980s expansion.
- Expansion and recession industries show similar rates conditional on macro time variation.

Economic message: The asset market is procyclical. Economy-wide activity is an important determinant of corporate asset transfers.

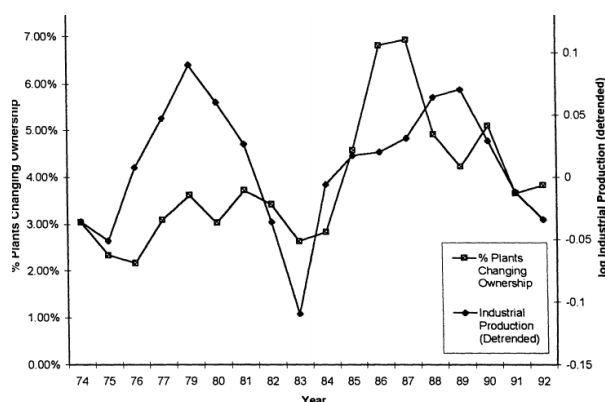


Figure 1. Left-hand scale is the total percentage of plants changing ownership by year. Right-hand scale is aggregate real industrial production detrended.

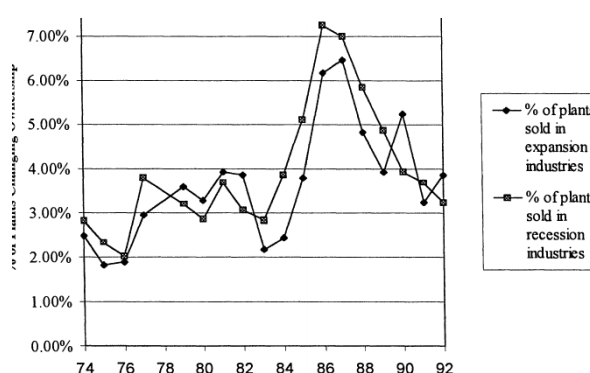


Figure 2. Change of ownership by type of industry. Expansion (recession) industries are industries in which real and detrended value of shipments at the industry level increases (decreases).

Figure 1 shows that, consistent with Table II, the annual percentage of plants that transferred ownership is high in expansions. It is highest in the fastest growing quartile of industries. However, the overall rate of transfers is also high in recession industries. (a) Figure 1: percentage of plants changing ownership by year. (b) Figure 2: change of ownership by type of industry.

7.4 Figure 3 and Figure 4: Transaction type and segment rank

Read: Figure 3 decomposes reallocations into mergers, full-segment sales, and partial-segment sales. Figure 4 shows asset-sale rates by segment rank.

- Mergers are strongly procyclical.
- Partial-firm sales are less volatile but important.
- Firms are much more likely to sell peripheral segments than main segments.

Economic message: Segment rank is a key indicator of comparative advantage. Firms keep core divisions and sell peripheral ones more often.



Figure 3. Percentage of plants reallocated through mergers, full-segment sales, and partial-segment sales by year.

(a) Figure 3: plants reallocated through mergers, full-segment sales, and partial-segment sales.

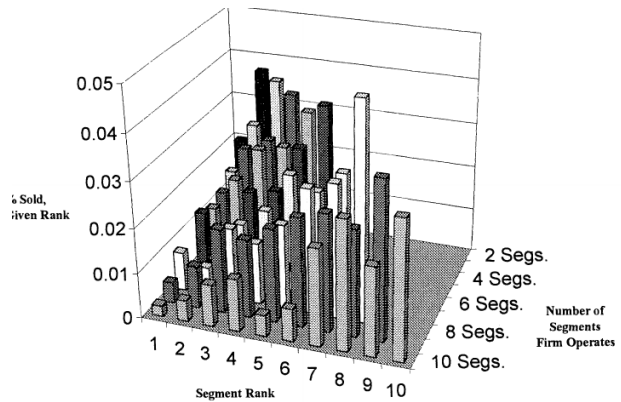


Figure 4. Percentage of plants sold via asset sale by segment rank as the number of segments the firm operates increases. Segment rank equals one for the largest segment a firm operates.

(b) Figure 4: plants sold via asset sale by segment rank.

7.5 Table IV and Table V: Probability of partial-firm asset sales

Read: Table IV estimates partial-firm sale probabilities using TFP. Table V shows robustness using value added per worker and cash flow.

- Lower plant productivity predicts higher sale probability.
- Lower segment productivity predicts higher sale probability for conglomerates.
- Aggregate industrial production increases sale probability.
- Alternative efficiency measures yield similar conclusions.

Economic message: Firms sell weak assets and weak segments, particularly in periods when demand makes asset redeployment more valuable.

Partial Firm Sales

Regressions test the effects of plant-level productivity and industry-level demand on the decision to sell plants for single-segment and multiple-segment firms. The dependent variable is a binary variable that indicates that the firm sells a plant, while still remaining in operation the next year. Observations are at the plant level for all firms. Detrended aggregate industrial production is actual minus predicted U.S. aggregate industrial production where predicted industrial prediction is obtained by regressing industrial production on a yearly time trend. Productivity variables are industry and year adjusted. Segment variables are aggregated into three-digit SIC codes for industry segments from underlying plant-level data. We estimate the regressions using unbalanced panel probit regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity consistent standard errors. Data are yearly from 1979 to 1992. (p -values are in parentheses.)

Variable	Dependent Variable: Plant Sale		Test for Significant Diff.: Single vs. Conglomerate Firms (p -Value) ^a
	Single segment firms (with >1 Plant)	Multiple segment firms	
Constant	-6.162 (0.000)*	-1.774 (0.000)*	(0.000)*
Aggregate industrial production (detrended)	0.831 (0.000)*	1.104 (0.000)*	(0.000)*
Industry sensitivity to aggregate industrial production	0.009 (0.277)	-0.006 (0.003)*	(0.056)***
Lagged productivity (TFP) of plant ^b	-0.021 (0.797)	-0.075 (0.001)*	(0.042)**
Lagged segment TFP	0.026 (0.705)	-0.338 (0.000)*	(0.000)*
Segment TFP * change in downstream industry demand	-0.055 (0.317)	0.000 (0.987)	(0.812)
Lagged firm size (coefficient * 10,000,000)	0.642 (0.000)*	-0.038 (0.000)*	(0.000)*
Average number of plants per industry segment	-0.002 (0.378)	-0.022 (0.000)*	(0.000)*
Firm multiple segment variables		0.080 (0.000)*	
Segment rank/number of segments		0.240 (0.000)*	
Other segment's weighted TFP ^c		0.073 (0.003)*	
Relative demand * other segments weighted TFP ^d			
Total plant years	45,959	259,065	
Chi-squared statistic	108.44	1940.91	
Significance level (p -value)	<1%	<1%	

^a, **, *** Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

^b Significance test for a multiple-segment dummy variable interacted with each independent variable in a regression with all firms with greater than one plant.

^c Total Factor Productivity (TFP) is calculated using a translog production function.

^d Other segments' productivity is weighted average of the firm's other segment(s) weighted by the segment(s) sales.

^e Relative industry demand equals one (zero, minus one) when the segment's change in downstream industry demand is greater (equal, less) than the change in downstream demand of the firm's median segment. Relative industry demand is interacted with the firm's other segments' weighted TFP.

(a) Table IV: partial-firm sales regressions.

Robustness Tests: Alternative Measures of Efficiency

Regressions test the effects of plant-level efficiency measures and industry-level demand on the decision to buy out another firm for single-segment and multiple-segment firms. The dependent variable is a binary variable which indicates that the firm sells a plant, while still remaining in operation the next year. Observations are at the plant level for all firms. Efficiency variables are industry and year adjusted. Segment variables are aggregated into three-digit SIC codes. Detrended aggregate industrial production is actual minus predicted U.S. aggregate industrial production where predicted industrial prediction is obtained by regressing industrial production on a yearly time trend. Columns one and two use value added per worker as the measure of efficiency. Value added is sales less materials divided by the number of workers. Columns three and four use plant cash flow as the measure of efficiency. Cash flow is sales less materials cost of goods sold less capital expenditures divided by sales. We estimate the regressions using unbalanced panel probit regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity consistent standard errors. Coefficients for the value added measure of efficiency are multiplied by one hundred. (p -values are in parentheses.)

Variable	Dependent Variable: Plant Sale			
	Efficiency measure: Value added per worker		Efficiency measure: Plant cash flow	
	Single-segment Firms (with >1 Plant)	Multiple-segment Firms	Single-segment Firms (with >1 Plant)	Multiple-segment Firms
Constant	-6.364 (0.000)*	-6.236 (0.000)*	-6.004 (0.000)*	-1.799 (0.000)*
Aggregate industrial production (detrended)	0.880 (0.000)*	0.980 (0.000)*	0.802 (0.000)*	1.100 (0.000)*
Lagged plant efficiency measure	-0.166 (0.000)*	-0.184 (0.000)*	-0.404 (0.003)*	-0.363 (0.000)*
Lagged segment efficiency measure	-0.004 (0.597)	-0.001 (0.670)*	0.345 (0.040)**	-0.162 (0.015)**
Lagged segment efficiency * change in downstream industry demand	0.061 (0.661)	0.046 (0.633)	-0.449 (0.147)	0.325 (0.748)
Lagged firm size (coefficient * 10,000,000)	0.710 (0.000)*	0.066 (0.000)*	0.671 (0.000)*	-0.037 (0.000)*
Average number of plants per industry segment	-0.003 (0.184)	-0.024 (0.000)*	-0.003 (0.224)	-0.022 (0.000)*
Firm multiple segment variables		0.489 (0.000)*		0.124 (0.000)*
Segment rank/number of segments		0.006 (0.792)		0.109 (0.097)***
Other segment's weighted efficiency ^a		0.129 (0.005)*		-0.040 (0.450)
Relative demand * other segments weighted efficiency ^b				
Total plant years	45,959	259,065	45,959	259,065
Chi-squared statistic	115.66	1561.24	95.39	623.74
Significance level (p -value)	<1%	<1%	<1%	<1%

^a, **, *** Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

^b Other segments' efficiency is weighted average of the firm's other segment(s) weighted by the segment(s) sales.

^c Relative industry demand equals one (zero, minus one) when the segment's change in downstream industry demand is greater (equal, less) than the change in downstream demand of the firm's median segment. Relative industry demand is interacted with the firm's other segments' weighted TFP.

(b) Table V: robustness using alternative efficiency measures.

7.6 Table VI and Table VII: Economic significance and full-firm selling

Read: Table VI translates regression coefficients into predicted sale probabilities. Table VII estimates firm selloff probabilities.

- Moving from high to low productivity meaningfully increases predicted sale probabilities.
- Demand conditions strongly change predicted probabilities.
- Whole-firm selloffs are also cyclical and related to productivity.

Economic message: The effects are economically meaningful, not only statistically significant. Both partial asset sales and whole-firm sales respond to productivity and demand.

TABLE VI
Probability of an Asset Sale

Predicted probability of an asset sale varying performance measures and industrial production using the regression coefficients from Tables IV and V. Panel A presents results varying performance measures from the 10th to 90th percentiles. Panel B presents results varying detrended industrial production from its median value using yearly data from 1979 to 1992. Panel C varies both productivity and aggregate industrial production at their median values. We hold all other variables at their sample medians.

Panel A: Varying Initial Productivity/Performance				
	10th Percentile	Median Level	90th Percentile	
Single-segment firms				
Varying productivity (Table IV, column 1)	1.08%	0.85%	0.66%	
Varying value added per worker (Table V, column 1)	1.02%	0.88%	0.72%	
Varying cash flow (Table V, column 3)	0.99%	0.87%	0.77%	
Conglomerate firms				
Varying productivity (Table IV, column 2)	4.32%	3.71%	2.86%	
Also decreasing segment rank to 10% (main divisions)	4.11%	3.52%	2.71%	
Also increasing segment rank to 90% (peripheral divisions)	4.80%	4.13%	3.20%	
Varying value added per worker (Table V, column 2)	4.06%	3.66%	3.08%	
Varying cash flow (Table V, column 4)	4.19%	3.59%	3.04%	
Panel B: Varying Industrial Production				
	10th Percentile	Median Level	90th Percentile	
Single-segment firms				
(Table IV, column 1)	0.57%	0.85%	1.26%	
Conglomerate firms				
All divisions (Table IV, column 2)	3.20%	3.71%	4.03%	
Also decreasing segment rank to 10% (main divisions)	3.02%	3.52%	3.81%	
Also increasing segment rank to 90% (peripheral divisions)	3.64%	4.13%	4.56%	
All divisions (Table V, column 2)	3.21%	3.66%	4.09%	
All divisions (Table V, column 4)	3.15%	3.59%	4.02%	
Panel C: Varying Industrial Production and Productivity				
	Productivity 10th and Industrial Production 90th percentile	Mean Level	Productivity 90th and Industrial Production 10th percentile	
Single-segment firms				
Varying productivity (Table IV, column 1)	1.46%	0.85%	0.48%	
Varying value added per worker (Table V, column 1)	1.39%	0.88%	0.51%	
Varying cash flow (Table V, column 3)	1.32%	0.87%	0.56%	
Conglomerate firms				
All divisions (Table IV, column 2)	4.82%	3.71%	2.50%	
Also decreasing segment rank to 10% (main divisions)	4.59%	3.52%	2.36%	
Also increasing segment rank to 90% (peripheral divisions)	5.34%	4.13%	2.80%	
All divisions (Table V, column 2)	5.53%	3.66%	2.70%	
All divisions (Table V, column 4)	4.67%	3.59%	2.66%	

(a) Table VI: predicted probability of an asset sale.

Table VII
Mergers and Firm Selloffs: Selling Firm Characteristics

Regressions test the effects of plant-level productivity and industry-level demand on the decision to sell out or merge with another firm for single-segment and multiple-segment firms. The dependent variable is a binary variable that indicates that the firm sells to another firm. Productivity variables are industry and year adjusted. Firm-level variables are aggregated from plant-level data by weighting each plant by its real value of plant shipments. Industry-level variables are constructed for each firm using as weights the real value of that firm's industry production. We estimate the regressions using unbalanced panel probit regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity-consistent standard errors following Huber-White. (*p*-values are in parentheses.)

Variable	Dependent Variable: Firm Selloff			Test for Significant Diff.: Multiple-segment Interaction Variable (<i>p</i> -value)*
	Single-segment Firms with 1 Plant	Single-segment Firms with >1 Plant	Multiple- segment Firms	
Constant	-1.533 (0.000)	-1.987 (0.000)*	-1.808 (0.000)*	(0.644)
aggregate industrial production (detrended)	0.723 (0.000)*	0.716 (0.035)**	0.743 (0.006)*	(0.952)
industry sensitivity to aggregate industrial production	0.012 (0.000)*	0.013 (0.054)***	0.018 (0.002)*	(0.519)
agged productivity of firm (TFP) ^b	-0.023 (0.041)**	-0.018 (0.526)	-0.014 (0.549)	(0.600)
firm TFP × downstream industry shipments	0.181 (0.191)	0.070 (0.864)	-0.568 (0.028)**	(0.615)
agged firm size (coefficient × 10,000,000)	2.500 (0.000)*	0.299 (0.186)	-0.145 (0.042)**	(0.166)
dispersion (herfindahl) across industries			-0.223 (0.000)*	
total firm years	96,961	24,772	42,524	
chi-squared statistic	56.12	11.62	39.36	
significance level (<i>p</i> -value)	<1%	0.040	<1%	

*, **, *** Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.
Significance test for a multiple-segment dummy variable interacted with each independent variable in a regression with all firms with > 1 plant.
Total Factor Productivity (TFP) is calculated using a translog production function.

(b) Table VII: mergers and firm selloffs: selling-firm characteristics.

7.7 Table VIII and Table IX: Buyer characteristics and merger probabilities

Read: Table VIII studies buying firm characteristics. Table IX reports predicted probabilities of mergers and full-firm purchases.

- Productive firms are more likely to acquire assets.
- Conglomerate buyers can be focus-increasing or focus-decreasing.
- Predicted merger probabilities rise with buyer productivity and industry demand.

Economic message: Buyers are firms with productive use for assets, which supports efficient redeployment rather than random acquisition.

Mergers and Firm Selloffs: Buying Firm Characteristics

Regressions test the effects of plant-level productivity and industry-level demand on the decision to buy out another firm for single-segment and multiple-segment firms. The dependent variable is a binary variable that indicates that the firm buys another firm. Observations are at the firm level for all firms. Productivity variables are industry and year adjusted. Segment-level variables are aggregated from plant-level data by weighting each plant by its real value of plant shipments. Industry-level variables are constructed for each firm using as weights the real value of a firm's industry production. We estimate the regressions using unbalanced panel probit regressions allowing for correlated residuals within panel units. Focus-increasing (decreasing) purchases are transactions that result in the buying firm's sum of squared industry market shares (herfandahl) increasing (decreasing). Significance tests are conducted using heteroskedasticity-consistent standard errors following Huber-White. (p -values are in parentheses.)

Variable	Dependent Variable: Firm Purchase			
	Single-segment Firms		Multiple-segment Firms	
	With 1 Plant	With >1 Plant	Focus- increasing Purchase	Focus- decreasing Purchase
Constant	-2.381 (0.000)*	-2.094 (0.000)*	-2.122 (0.000)*	-1.987 (0.000)*
Aggregate industrial production (detrended)	0.551 (0.044)**	0.450 (0.206)	0.914 (0.001)*	0.817 (0.001)*
Industry sensitivity to aggregate industrial production	0.007 (0.216)	-0.008 (0.318)	-0.006 (0.371)	-0.004 (0.512)
Lagged productivity of industry segment (TFP) ^a	-0.048 (0.413)	0.271 (0.004)*	0.249 (0.100)***	0.290 (0.000)*
Firm TFP * downstream industry shipments	0.618 (0.580)	0.884 (0.643)	0.062 (0.549)	-0.047 (0.412)
Lagged firm size (coefficient * 10,000,000)	4.330 (0.000)*	0.650 (0.105)	0.196 (0.091)***	0.166 (0.051)**
Average number of plants per industry segment		0.027 (0.000)*	0.090 (0.000)*	0.075 (0.000)*
Total plant years	77,713	24,774	40,152	40,498
Chi-squared statistic	21.33	99.85	141.37	144.31
Significance level (p -value)	<1%	<1%	<1%	<1%

*, **, *** Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

^a Total Factor Productivity (TFP) is calculated using a translog production function.

(a) Table VIII: mergers and firm selloffs: buying-firm characteristics.

Probability of Mergers

Predicted probability of a merger or full-firm purchase varying performance measures and industrial production using the regression coefficients from Tables VII and VIII. Panel A presents results varying performance measures from the 10th to 90th percentiles. Panel B presents results varying detrended industrial production from its median value over 1979 to 1992. Panel C varies both productivity and aggregate industrial production at their median values. We hold all other variables at their sample medians.

Panel A: Selling Firms			
Varying productivity	10th Percentile	Median	90th Percentile
Single-segment selling firms	2.63%	2.55%	2.45%
Conglomerate selling firms	3.21%	3.01%	2.77%
Varying industrial production	10th Percentile	Median	90th Percentile
Single-segment selling firms	2.33%	2.55%	2.75%
Conglomerate selling firms	2.74%	3.01%	3.25%
Varying productivity & industrial production	Productivity 10th, Industrial Production 90th Percentile	Median Level	Productivity 90th, Industrial Production 10th Percentile
Single-segment selling firms	2.83%	2.55%	2.26%
Conglomerate selling firms	3.46%	3.01%	2.53%
Panel B: Buying Firms			
Varying productivity	10th Percentile	Median	90th Percentile
Single-segment buying firms	1.85%	2.13%	2.56%
Conglomerate buying firms: Focus increasing	2.30%	2.47%	2.94%
Conglomerate buying firms: Focus decreasing	2.92%	3.27%	3.95%
Varying industrial production	10th Percentile	Median	90th Percentile
Single-segment buying firms	1.93%	2.13%	2.17%
Conglomerate buying firms: Focus increasing	2.03%	2.47%	2.56%
Conglomerate buying firms: Focus decreasing	2.76%	3.27%	3.37%
Varying productivity & industrial production	Productivity 10th, Industrial Production 90th Percentile	Median Level	Productivity 90th, Industrial Production 10th Percentile
Single-segment buying firms	1.78%	2.13%	2.60%
Conglomerate buying firms: Focus increasing	2.22%	2.47%	3.04%
Conglomerate buying firms: Focus decreasing	2.87%	3.27%	4.01%

(b) Table IX: predicted probability of mergers and full-firm purchases.

7.8 Table X and Table XI: Productivity changes after transactions

Read: Table X summarizes productivity changes around asset sales and mergers. Table XI regresses productivity changes on buyer and seller characteristics.

- Peripheral divisions sold to more suitable buyers often improve productivity.
- Productivity gains are stronger when buyers are initially more productive than sellers.
- The improvement depends on the match between buyer and seller organization.

Economic message: The post-transaction evidence supports the view that asset markets can improve the allocation of resources.

Summary Statistics for Change in Productivity for Assets Sold or Merged

The change in productivity, industry and year adjusted, for three years surrounding asset sales, mergers, or full-firm sales (year -1 to year $+2$). Values represent sample means for transactions between buyers and sellers. (Standard error of the mean in parentheses.) Third row is the number of plants transacted.

Panel A: Partial-Firm Sales				
Selling Firm Characteristics				
	Multidivision Firm		Single-segment Firm	
	Main Division	Peripheral Division		
Buying firm characteristics				
Multidivision buyer adds to Main division	0.062 (0.039)	0.103* (0.036)	0.127 (0.092)	
Peripheral division	444 -0.008 (0.027)	580 0.047** (0.021)	87 0.033 (0.076)	
	857 0.019 (0.052)	1,544 0.107** (0.042)	122 -0.038 (0.069)	
Single-segment buyer	275	382	126	
Panel B: Full-Firm Mergers and Acquisitions				
Selling Firm Characteristics				
	Multidivision Firm		Single-segment Firm	
	Main Division	Peripheral Division	with 1 Plant	with >1 Plant
Buying firm characteristics				
Multidivision buyer adds to Main division	0.087** (0.041)	0.143*** (0.083)	-0.020 (0.039)	-0.039 (0.043)
Peripheral division	396 -0.017 (0.020)	89 -0.020 (0.025)	420 -0.051*** (0.028)	234 0.058 (0.039)
	1,278 -0.044 (0.049)	914 0.037 (0.067)	723 0.053 (0.029)	401 -0.011 (0.050)
Single-segment buyer	237	108	677	219

*, **, *** Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

(a) Table X: change in productivity for assets sold or merged.

Change in Productivity Based on Buying & Selling Firm Characteristics

This table presents a single regression that tests the joint effects of buyer and selling firm organization characteristics on the change in plant productivity over a three-year horizon from the year prior to the asset sale or merger (year -1) to the end of the second year (year $+2$) after the transaction. Total Factor Productivity (TFP) variables are calculated using a translog production function. Lagged firm productivity variables are aggregated up from individual plants. All productivity variables are industry and year adjusted. Transaction type variables are indicator variables that show whether the transaction was from a seller's main or peripheral division or from a single-segment firm to a buyer's main, peripheral, or to a single-segment buyer. Transaction type dummy variables are then interacted with the difference in productivity between the buying firm and the assets purchased. We estimate the regressions using ordinary least squares. Significance tests are conducted using heteroskedasticity-consistent standard errors. (p -values in parentheses.)

Variable	Dependent Variable: Change in Productivity Year -1 to $+2$	
	Transaction Type	
	Partial Firm Sale	Merger/Acquisition
Productivity and transaction size		
Lagged productivity (year -1)	-0.453 (0.000)*	
of selling firm's plants	0.399	
Difference in productivity (year -1)		
buying firm—productivity of plants purchased	(0.000)*	
Difference in productivity (year -1) * Indicator	-0.084	
variable for negative difference in productivity	(0.004)*	
Buyer size (coefficient * 10,000,000)	-0.003 (0.834)	
Seller size (coefficient * 10,000,000)	0.123 (0.000)*	
Difference in productivity interacted with transaction characteristics		
Sale from seller's main division	-0.009 (0.931)	0.000 (0.907)
to buyer's main division		
Sale from seller's main division	-0.014 (0.894)	-0.043 (0.676)
to buyer's peripheral division		
Sale from seller's main division	-0.032 (0.776)	-0.119 (0.282)
to a single segment buyer		
Sale from seller's peripheral division	0.119 (0.268)	omitted
to buyer's main division		
Sale from seller's peripheral division	-0.002 (0.987)	-0.058 (0.572)
to buyer's peripheral division		
Sale from seller's peripheral division	-0.090 (0.490)	-0.165 (0.179)
to a single segment buyer		
Sale from single segment firm	0.021 (0.883)	-0.034 (0.750)
to buyer's main division		
Sale from single segment firm	0.065 (0.620)	-0.045 (0.662)
to buyer's peripheral division		
Sale from single segment firm	-0.034 (0.786)	-0.155 (0.131)
to a single segment buyer		
Constant		-0.066 (0.000)*
Total transactions:		
Partial firm sales and merger and acquisitions	10,109	
R^2	0.21	
F-test all coefficients = 0 (p -value)	<1%	

(b) Table XI: productivity changes by buyer and seller characteristics.

7.9 Table XII and Appendix Figure A1: Buyer productivity and TFP measurement

Read: Table XII asks whether buyers improve productivity after acquiring assets. Figure A1 shows the distribution of R^2 values from within regressions used in TFP estimation.

- Buyer's existing assets can improve after acquisitions, especially when the buyer was initially less productive than purchased assets.
- Figure A1 supports the reliability of plant-level TFP measurement.

Economic message: Acquisitions can create synergies for the buyer as well as the acquired plant; the appendix supports the productivity measurement behind these tests.

Change in Buyer Productivity

Buyer's change in productivity, industry and year adjusted, for three years surrounding asset purchases, mergers, or acquisitions (year -1 to year +2). Values represent sample means for transactions. We construct a firm-level weighted average of buyer's plants, with weights equal to an individual plant's value of shipments. For the existing assets' change in productivity, we present the weighted average change in productivity for plants in the same industry as the purchased assets. (Standard error of the mean is in parentheses.)

	Overall Change in Productivity	Cases where	
		Initial Buyer Productivity > Productivity of Assets Purchased	Initial Buyer Productivity < Productivity of Assets Purchased
Panel A: Change in Productivity of Buyer's Existing Assets (Excluding Purchased Assets)			
All transactions	0.004 (0.009)	-0.027** (0.011)	0.053* (0.015)
Asset purchases	0.008 (0.014)	-0.028*** (0.016)	0.065* (0.023)
Mergers & acquisitions	0.000 (0.012)	-0.027*** (0.016)	0.044** (0.019)
Panel B: Change in Productivity of All Assets of Buyer (Including Purchased Assets)			
All transactions	0.020* (0.006)	0.028* (0.008)	-0.010 (0.011)
Asset purchases	0.048* (0.009)	0.056* (0.010)	0.002 (0.014)
Mergers & acquisitions	-0.003 (0.007)	0.004 (0.009)	-0.028** (0.013)
Number of buying firms	6905	5366	1539

*, **, *** Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

(a) Table XII: change in buyer productivity.

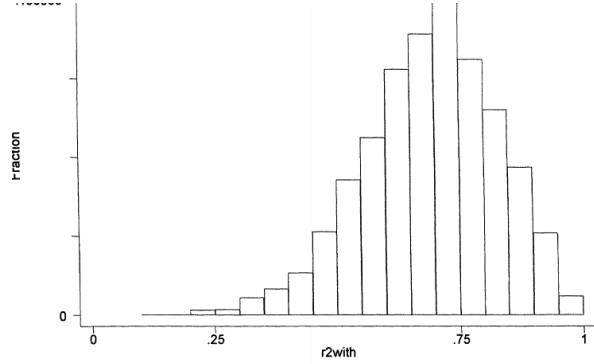


Figure A1. Histogram of R -squares from TFP within regressions run each year using five years of lagged data.

(b) Figure A1: histogram of R^2 values from TFP regressions.

7.10 Appendix Table A1: Total Factor Productivity

Read: Table A1 reports coefficients from production-function regressions used to construct TFP.

Economic message: The appendix table documents the production-function foundation for the paper's central efficiency measure.

Total Factor Productivity

Table reports mean coefficients from yearly regressions by industry. Each regression is repeated in each year for each industry using the last five years of data. Coefficients in column one represent the average coefficients from regressions estimated in each year for each industry (1,726 total regressions). Columns two and three represent illustrative industries for 1978. In each year, productivity is the residual plus the firm-specific fixed effect obtained from regressing the real value (deflated) of output on a plant's age, the value of materials and inputs, the real value of a plant's capital stock and the hours of labor utilized. Adjustments for depreciation for capital stock are made each year using depreciation deflators from the Bureau of Economic Analysis. All dollar amounts are deflated to 1982 dollars using deflators for the National Bureau of Economic Research. (*p*-values in parentheses).

Independent Variable: Real Value of Output			
Variable	Mean Coefficient	Meat Packing SIC 2011	Oil Refining SIC 2911
Age	0.013 (0.061)**	0.019 (0.587)	0.169 (0.009)*
Inputs	0.108 (0.000)*	0.003 (0.972)	1.661 (0.000)*
Capital stock	0.044 (0.000)*	0.084 (0.124)	0.306 (0.001)*
Labor	0.674 (0.000)*	0.773 (0.000)*	-0.072 (0.600)
(Inputs) ²	0.074 (0.000)*	-0.001 (0.825)	0.009 (0.043)**
(Capital stock) ²	0.000 (0.631)	0.075 (0.000)*	0.007 (0.411)
(Labor) ²	0.020 (0.000)*	0.085 (0.000)*	-0.024 (0.039)**
Inputs * capital stock	-0.029 (0.000)*	-0.005 (0.640)	0.003 (0.856)
Inputs * labor	-0.110 (0.000)*	-0.005 (0.503)	-0.040 (0.003)*
Capital stock * labor	0.044 (0.000)*	-0.144 (0.000)*	-0.000 (0.999)
Constant	3.352 (0.000)*	2.819 (0.000)*	-3.765 (0.000)*
R^2		0.92	0.74

*, **, *** For column one mean value of the coefficient is significantly different from zero at the 10%, 5%, and 1% level, respectively.

Table A1: total factor productivity regressions. Single visual is set to width 0.6\linewidth by the notes rule.

8 Short summary

- The paper studies the market for corporate assets using plant-level Census data.
- Roughly 3.9 percent of plants change ownership annually from 1974 to 1992.
- Asset reallocations are strongly procyclical.
- Mergers, full-segment sales, and partial-segment sales are all important transaction types.
- Firms are more likely to sell peripheral divisions than main divisions.
- Low-productivity plants and segments are more likely to be sold.
- Productive firms are more likely to buy assets.
- Transacted plants often improve productivity when they move to better-suited owners.
- The evidence supports an efficient reallocation interpretation of many corporate asset transactions.

9 Conclusion

9.1 Core conclusion

Maksimovic and Phillips show that the corporate asset market plays an important reallocative role. Plant-level evidence indicates that many asset sales and mergers move productive assets toward firms that can operate them more efficiently.

9.2 Incremental takeaway

The paper’s contribution is to move merger analysis from firm-level event studies to plant-level real outcomes. This allows the authors to test whether the physical assets being traded become more productive, which is a direct test of resource reallocation.

9.3 Implications for corporate finance

The findings temper purely agency-based interpretations of mergers and diversification. Some transactions may be inefficient, but the systematic evidence shows that asset markets often operate in line with profit-maximizing reallocation.

9.4 Implications for industrial organization

The results show that firm boundaries are endogenous and respond to productivity and demand. Firms buy and sell assets as their comparative advantage changes across industries and over the business cycle.

9.5 Limitations

The paper cannot fully observe all motivations behind transactions and relies on manufacturing plants, so external validity to services or intangible assets is not automatic. Selection into transactions remains important, even though post-transaction productivity changes provide strong supporting evidence.

9.6 Takeaway

Corporate asset sales and mergers should be viewed not only as financial events but also as real reallocations of productive capital. The market for corporate assets helps reassign plants to owners with better fit, especially when demand conditions make redeployment valuable.